D4.3 Climate change

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d4-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 820–833 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

CAUSES AND FEEDBACK

D4.3.1 Human causes of climate change

pp. 820–824

Human activity has raised atmospheric **carbon dioxide**, mainly through burning fossil fuels and deforestation, and **methane**, for example from ruminants, rice fields and landfill. These gases absorb long-wave radiation and warm the atmosphere.

D4.3.2 Positive feedback

pp. 825–827

In a positive feedback, warming causes changes that lead to more warming. Examples are carbon dioxide released from the deep ocean, more solar radiation absorbed as snow and ice are lost, faster decomposition in thawing permafrost, and more droughts and forest fires.

Watch out: positive does not mean good. It means the effect increases the original change.

D4.3.3 Boreal forests as a tipping point

pp. 827–828

Warmer temperatures and less winter snow bring more drought, and primary production falls in the taiga. Forests turn brown and die back, so they may change from accumulating carbon to losing it.

EFFECTS ON ECOSYSTEMS

D4.3.4 Melting ice changes polar habitats

p. 829

Early break-up of Antarctic **landfast ice** threatens the breeding grounds of emperor penguins. Loss of Arctic **sea ice** removes the resting places of Pacific walruses.

D4.3.5 Ocean currents and upwelling

pp. 829–830

Warmer surface water can prevent upwelling of nutrients. This changes its timing and extent, reducing primary production in the ocean and the flow of energy through marine food chains.

D4.3.6 Range shifts

pp. 830–831

Species are moving **upslope** and **towards the poles**. On two mountains in Papua New Guinea, the upper limits of bird species moved up by about 110 to 150 m between the 1960s and 2012–13. Ranges of tree species in North America are contracting in the south and spreading north.

D4.3.7 Threats to coral reefs

p. 832

Threats to coral reefs

D4.3.7



Bleached coral. Acropora, CC BY-SA 3.0.



A healthy reef. Jim Maragos, public domain.

Threat	Cause	Effect
Acidification	More CO_2 dissolves; pH falls	Slower calcium carbonate
Bleaching	Warmer water stresses corals	Algae expelled; corals may die

Collapse
Reefs hold huge diversity; losing corals removes habitat and food for many species.

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More dissolved carbon dioxide causes **ocean acidification**, which suppresses calcification. Warmer water causes **bleaching**, as corals expel their algae. Reefs cover less than 0.1% of the ocean yet hold about a quarter of marine species, so their loss could cause the ecosystem to collapse.

RESPONSE

D4.3.8 Capturing carbon

p. 833

Approaches include **afforestation**, **forest regeneration** and **restoring peat-forming wetlands**. Hinewai Reserve in New Zealand is an example of rapid regeneration after degradation. Use a local peatland as a second example.

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